AIP304 Machine Learning – Coursework 2

Plant disease recognition in the wild

1 Overview

In this coursework, you will build and evaluate a machine learning system that recognises plant diseases from photographs taken in varied real-world conditions. You will use a dataset, implement your system in PyTorch, and explain what your results show.

Complete one of two options. Option 1 is the standard task: classify images with a convolutional neural network (CNN). Option 2 is an optional research challenge: recognise known classes while rejecting unfamiliar host–disease combinations. Both options use the same dataset subset and can earn the full 100 marks.

Assessment detail	Requirement
Release	Week 8 – Monday, 2 November
Deadline	Week 13 – Sunday, 13 December, 23:59
Group size	1–3 students
Submission	One Jupyter notebook, one PDF report of at most 5 pages, and one video of at most 3 minutes
Environment	Python and PyTorch; runnable in Google Colab or Kaggle
Total mark	100 points across the three submission items

1.1 Learning outcomes

B. Evaluate and compare the key properties and characteristics of existing and emerging machine learning algorithms.
C. Select and apply appropriate machine learning algorithms to solve specific computational or real-world problems.
D. Critically adapt and optimise machine learning algorithms to meet defined performance or application needs.

1.2 General expectations

Work *individually* or in *a team of up to three*. Submit one package per team and identify all members in the notebook and report. Pretrained networks and transfer learning are allowed. All members must understand the submitted work. The report must disclose AI use and, for teams, include a contribution statement.

Your notebook must generate the results and visualisations used in the report. The video must demonstrate working inference and show the presenter's face during narration. Every team member must appear. Detailed submission guidance appears in Section 6.

1.3 Late or incomplete submission

Late work is accepted for up to five working days. Each commenced 24-hour period incurs a deduction of 5% of the total available marks, equivalent to five points out of 100 per day. For example, 80 becomes 70 after two late days. Marks cannot fall below zero. Submissions *more than five days late* or *incomplete submissions* receive **zero marks**.

2 Dataset and experimental requirements

2.1 Dataset and coursework subset

Use PlantSeg version v7, released on 25 November 2025 (Wei et al., 2025). Download plantseg.zip, approximately 1.1 GB, from the versioned record below. The 2026 data descriptor reports 7,774 images, 34 plant hosts, and 115 host–disease categories. Images have disease-region masks and host/disease metadata (Wei et al., 2026).

Download: https://zenodo.org/records/17719108

Retain images from these nine hosts: apple, banana, citrus, corn, potato, soybean, tomato, wheat, and zucchini. This selection gives 43 host–disease categories in the published taxonomy. Derive actual counts and class identifiers from the downloaded metadata. Do not mix dataset releases.

Treat each host–disease pair as one class. For example, potato late blight and tomato late blight are separate classes. Preserve the supplied labels and acknowledge dataset and image sources. Option 1 uses RGB images and image-level labels. Option 2 compares a method using binary lesion-mask supervision from known training classes with a matched baseline trained without that supervision. All evaluated predictions must use only RGB images.

2.2 Training validation and test protocol

Preserve the supplied approximately 70% training, 10% validation, and 20% test partition when selecting the subset. Option 2 additionally assigns classes to known and unknown roles. Record a fixed manifest of image identifiers, labels, and partition membership.

Partition	Permitted use
Training	Learn model parameters; estimate class weights and any dataset-specific preprocessing statistics.
Validation	Choose architecture, augmentation, hyperparameters, stopping epoch, calibration, and decision thresholds.
Test	Evaluate frozen configurations and carry out final error analysis. Do not tune from test results.

1. **Audit the data.** Report class counts per partition, unreadable files, and duplicate checks. Keep identical images within one partition; document exclusions before modelling and use the same manifest for all methods. Flag possible near-duplicate leakage as a limitation where unresolved.
2. **Prevent leakage.** Keep augmented copies with their original training image. Fit preprocessing on training data only, or use the documented transforms supplied with pretrained weights. Never use class-bearing filenames or metadata as predictive inputs.
3. **Evaluate consistently.** Use deterministic validation/test transforms, evaluation mode, and disabled gradients. Select checkpoints using validation results; freeze all planned baselines and ablations before final testing.
4. **Record reproducibility details.** Document random seeds, software versions, pretrained-weight identifiers, hardware, training time, and checkpoint-selection rules.

2.3 Practical implementation

A small CNN or an ImageNet-pretrained ResNet-18 or MobileNet is suitable. Start near 224 × 224 pixels with a modest training budget. Justify resizing and cropping so that symptoms are not systematically removed. Generic pretrained weights are allowed; do not use checkpoints trained on PlantSeg or overlapping PlantWild images. The PyTorch transfer-learning tutorial is a useful starting point (Chilamkurthy, 2025).

3 Option 1: CNN image classification

Objective: Predict the host–disease class of a single RGB photograph from the coursework subset.

This is closed-set classification: test photographs are unseen, but their possible categories are represented during training. Figure 1 shows how labelled training images support prediction of a new image.

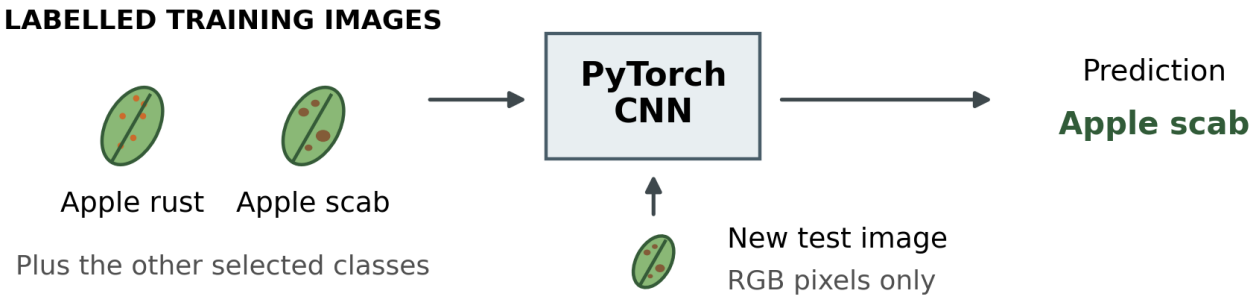


Figure 1. Classification using labelled plant images. The leaves and prediction are schematic illustrations, not dataset photographs or measured results.

3.1 Required tasks

1. **Explore the data.** Display representative training images and class frequencies. Explain challenges such as similar symptoms, small lesions, varying backgrounds, and imbalance.
2. **Build baselines.** Implement a majority-class predictor and a basic CNN system. The CNN may start from random weights or a pretrained feature extractor.
3. **Improve the CNN.** Explain architecture, loss, optimiser, learning rate, augmentation, and stopping. Compare at least two controlled changes, such as fine-tuning, class weighting, or augmentation. Change one factor at a time where possible.
4. **Evaluate and explain.** Select the final configuration using validation macro-F1. Report test results for the baseline and improved systems, and analyse representative successes and failures. A small set of well-explained experiments is sufficient.

3.2 Required evaluation metrics

Metric	Meaning
Macro-F1 Primary metric	Compute F1 for each class, then average equally across classes. This gives rare classes equal importance.
Top-1 accuracy	Number of correctly classified images divided by the number evaluated. It may hide poor minority-class performance.
Balanced accuracy	Mean recall across classes: the average proportion of each class correctly recognised.
Per-class results	Precision, recall, F1, and support (number of examples), together with a confusion matrix.

$$P_c = \frac{TP_c}{TP_c + FP_c}\ ;\quad R_c = \frac{TP_c}{TP_c + FN_c}\ ;\quad F1_c = \frac{2TP_c}{2TP_c + FP_c + FN_c}$$

Here TP, FP, and FN denote true positives, false positives, and false negatives for class c. Macro-F1 is the mean of class-wise F1 values, not F1 calculated from averaged precision and recall. Use a fixed class order; state how undefined values and classes with no evaluation support are handled.

4 Option 2: Research challenge

4.1 Question and motivation

Can binary lesion-mask supervision during training improve rejection of held-out disease classes on familiar plant hosts, compared with an otherwise matched classifier trained without lesion masks, while maintaining known-class macro-F1 when inference uses only RGB images? If you have prior experience in Deep Learning research, do try this optional challenge!

Unknown plant-disease recognition is an established research area (Dong et al., 2024). Related work includes lesion-focused attention for open-set crop-disease recognition (Yan et al., 2026), zero-shot identification of unseen crop–disease pairs (Chai et al., 2024), and grounded recognition with healthy-image abstention (Alshehhi et al., 2026). These studies motivate investigating the specific contribution of binary lesion supervision under the coursework protocol.

Investigate whether lesion supervision provides a measurable benefit on the specified PlantSeg subset and class assignments. Use a current literature search to determine whether your study is a replication, an extension, or a new comparison. Explain its relationship to prior work and support any novelty claim with evidence.

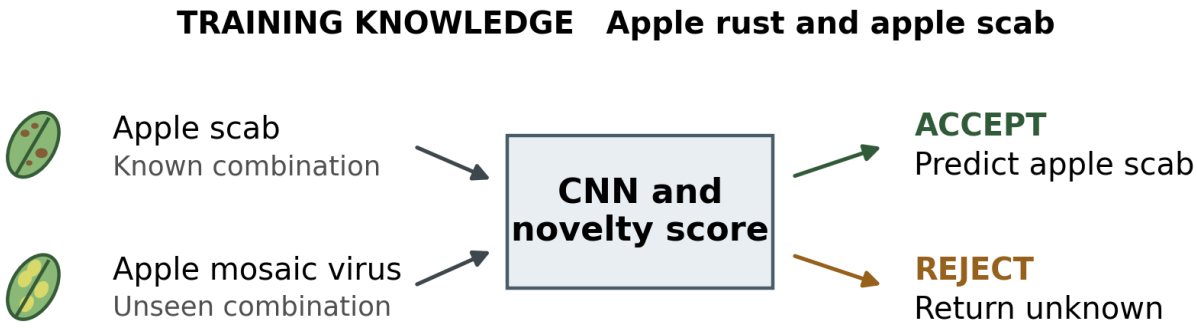


Figure 2. Desired behaviour for the illustrated apple split. Both inputs show a familiar host; one combination is known and one is unseen. Schematic leaves do not represent measured results.

Unknown means a host–disease combination absent from training. The disease type itself may be known on another host.

4.2 Construct the research split

1. **Reserve disease classes on familiar hosts.** Use all citrus classes as development unknowns and all potato classes as test unknowns.
2. **Reserve unfamiliar combinations.** For each of the other seven hosts, sort its disease labels alphabetically. Reserve the first category as a development unknown and the second as a test unknown. The remaining categories are known classes.
3. **Freeze the mapping.** Save the complete class-role mapping before training. For apple, black rot is the development unknown and mosaic virus is the test unknown; rust and scab are known classes.

Experimental set	Images permitted from the original partition
Training	Training images of known classes only.
Validation	Validation images of known classes and development-unknown classes.
Test	Test images of known classes and test-unknown classes.

Discard unused images of reserved classes. Development unknowns support validation only. No test-unknown images may enter training, feature fitting, calibration, or model selection. Train each class

assignment independently from generic initialisation; do not reuse weights from Option 1 or another assignment.

Evaluate held-out disease classes on familiar hosts separately from unfamiliar-host controls. The familiar-host evaluation directly addresses the research question; a pooled score alone cannot establish success.

For each class assignment, record whether each familiar-host unknown class has a disease label that also occurs among known classes on another host. Where sample sizes permit, report results separately for shared and previously unseen disease labels. Use the dataset's labels consistently and document any label normalisation.

Treat unfamiliar-host results as a supplementary control. Since the prescribed test control contains only potato, avoid general claims about performance across unfamiliar plant hosts.

5. Option 2 Research requirements and evaluation

5.1 Required investigation

1. Review current literature. Discuss at least three closely related methods, including work on unknown plant-disease recognition and lesion-guided or grounded recognition. Distinguish rejecting an unknown class from predicting the identity of an unseen class or abstaining on a healthy image. Compare the relevant methods' supervision, training data, inference inputs, and evaluation protocols. Explain whether your investigation replicates, extends, or provides a new comparison with prior work.

2. State a hypothesis. Before training, state a falsifiable expectation about the effect of lesion supervision. For example, predict that it will increase familiar-host AUOSCR while reducing known-class macro-F1 by no more than a predefined tolerance. State and justify that tolerance before examining results, and use it consistently when interpreting both validation and test performance. Define what evidence would support, contradict, or leave the hypothesis unresolved.

3. Develop one focused method. Investigate a mechanism through which binary lesion supervision could affect the learned representation, such as auxiliary lesion prediction, lesion-aware feature pooling, or consistency between global and lesion-focused features. Explain why the mechanism might improve unknown rejection and why it might fail. An adaptation of an established method is acceptable when it enables a clear and informative investigation. A backbone replacement alone does not test the effect of lesion supervision.

4. Compare meaningful baselines. Include an otherwise matched classifier trained without lesion supervision. Control the backbone, initialisation source, image preprocessing, augmentation, training budget, and hyperparameter-search budget. Evaluate maximum-softmax, maximum-logit, and energy-based rejection scores using the same trained classifier where applicable (Vaze et al., 2022; Liu et al., 2020). For the main comparison, use the same rejection-scoring rule for the baseline and lesion-supervised models. Reproduce or adapt a closely related method where feasible, and explain any differences in supervision, data, or computational resources.

5. Run ablations and repeated trials. Remove lesion supervision and any additional component claimed to contribute to the result. Repeat the main comparison using three training seeds for each of two predefined class assignments: six runs per model configuration. Assignment A is the mapping in Section 4.2. For Assignment B, retain the citrus and potato roles; for each of the other seven hosts, reserve the second alphabetically ordered disease category as a development unknown and the third as a test unknown, with all remaining categories known. Freeze both mappings and the evaluation protocol before training. Train each assignment independently without reusing weights fine-tuned on another assignment. Report seed means, standard deviations, and paired differences between methods separately for each assignment.

Use binary lesion/background masks from known training classes only within the training procedure. Class-coded masks must not be predictive inputs. Every validation and test prediction must use only

RGB images, without ground-truth masks or host metadata. A model may predict its own lesion map from RGB input.

Explain whether the evidence supports the hypothesis, including variation across seeds, class assignments, and unknown groups. Mixed and negative findings are valid outcomes. If additional time is available, a control using shuffled masks from known training images can help examine whether any benefit depends on meaningful image–mask alignment.

5.2 Required metrics

Let s(x) be a score where larger values mean stronger evidence of a known class. Accept an image when s(x) is at least the threshold. The primary research outcome is AUOSCR calculated using known test images and familiar-host test-unknown images. Report the corresponding evaluation with unfamiliar-host test unknowns separately, using the same known test images as the reference group. Report known-class macro-F1 alongside AUOSCR to show whether any improvement in rejection is accompanied by a loss of known-class recognition.

Metric	Required calculation and interpretation
Known-class performance	Macro-F1 and accuracy on known test images before rejection, using Section 3 definitions.
Unknown AUROC	Area under the ROC curve for unknown versus known images. Unknown is positive and −s(x) supplies continuous scores. A value of 0.5 means chance ranking; 1 means perfect ranking.
Area under the OSCR curve (AUOSCR)	Sweep the acceptance threshold. Plot correct-and-accepted known-image rate against unknown false-acceptance rate; integrate over the full false-acceptance range. Higher is better.
Fixed operating point	Choose a threshold accepting approximately 95% of known validation images. Report test known acceptance, correct-and-accepted known rate, and unknown false-acceptance rate.

Select each training run's checkpoint using known-class validation macro-F1. Select model configurations using familiar-host validation AUOSCR, subject to the predefined acceptable reduction in known-class macro-F1 relative to the matched baseline. Choose the primary rejection-scoring rule using baseline validation results and apply that same rule to both models in the main comparison. Retain the required comparisons with the other scoring rules.
Freeze all choices for both class assignments before examining either assignment's test results. Set the fixed operating-point threshold using known validation images only, as specified above. Assess the hypothesis using AUOSCR differences, known-class macro-F1 differences, and variation across seeds and assignments; distinguish a consistent improvement from a small or inconsistent numerical difference.

$$CCR(\tau) = \frac{\text{Number of known images correctly classified and accepted}}{\text{Number of known images}}$$

$$FPR_U(\tau) = \frac{\text{Number of unknown images accepted as known}}{\text{Number of unknown images}}$$

OSCR (open-set classification rate) evaluates classification and rejection together (Dhamija et al., 2018). Report seed means, standard deviations, and group sizes. Set thresholds on validation only; test known acceptance may differ from 95%.

6 Submission and communication guidance

6.1 Jupyter notebook

Submit an .ipynb file that runs in a fresh **Google Colab or Kaggle** environment without private paths or credentials. Include setup and dependencies; dataset access and checksum checks; split construction; exploration; baseline and model training; validation; final evaluation; figure generation; and an inference demonstration.

Keep the notebook readable from top to bottom, with explanatory Markdown and saved outputs. Include full training code. If checkpoints provide faster reproduction, include accessible checkpoints and loading instructions with the notebook submission. Document full-experiment and short-demonstration modes, hardware, and approximate runtime.

6.2 Concise report

Submit one PDF of at most **five pages**. The limit includes figures, tables, references, AI disclosure, and contribution statements; do *not* add a separate cover or appendix. Use readable typography and numbered figures and tables. Identify the selected option and all student names and identifiers.

Recommended structure: problem and motivation; related work; data and protocol; method; experiments and results; interpretation and limitations; references and disclosures. Explain the decisions needed to understand and reproduce your claims. For Option 2, explicitly include the current literature search, closest methods, hypothesis, baselines, and ablations.

6.3 Visual evidence

The notebook must regenerate every data-derived figure and results table used in the report. Use a small number of informative visuals; possible elements include:

- Class-frequency bars with labelled categories and counts to show imbalance.
- Training and validation curves against epoch, with the selected checkpoint marked.
- A confusion heatmap with true labels on rows, predicted labels on columns, and a labelled colour scale. A readable subset may appear in the report; retain the full matrix in the notebook.
- A prediction gallery with true class, predicted class, and model score, including failures.
- For Option 2, OSCR curves, known/unknown score distributions, and ablation plots with error bars.

Give every visual a useful caption, clear labels, and sample counts where relevant.

6.4 Video demonstration

Submit a video of at most three minutes. Explain the problem, show working inference, present one important result, and discuss one limitation. The presenter’s face must be visible during narration; picture-in-picture is acceptable. State the selected option and identify the team.

6.5 AI disclosure and team contributions

The report must include an AI-use disclosure naming any AI tools, versions where available, tasks assisted, and how outputs were checked. Cover assistance with code, writing, literature search, analysis, and visualisation. If no AI tools were used, state this explicitly. You remain responsible for correctness, originality, citations, and reproducibility.

Teams must include a contribution statement identifying each member’s work on implementation, experiments, analysis, writing, and demonstration. All members must understand the results. Selected teams may undergo further verification interview.

7 Grading rubric

The notebook, report, and video are assessed together against the criteria below.

Criterion	Points	Evidence for full credit
Problem framing and understanding	10	Clear objective and justified approach. Option 2 also gives a current literature search, closest-method comparison, and explicit hypothesis.
Data and experimental protocol	15	Correct labels and partitions, useful audit, leakage prevention, appropriate preprocessing, and transparent exclusions. Option 2 separates known and unknown roles correctly.
Implementation and method	20	Working PyTorch system, correct training and evaluation behaviour, justified choices, and faithful implementation of the claimed method.
Experimental comparisons	20	Meaningful baselines and controlled changes. Option 2 includes ablations, repeated trials, and an alternative class assignment. Comparisons use fair resources and consistent protocols.
Evaluation and analysis	15	Correct metrics, informative visualisations, convincing interpretation, representative failure analysis, and conclusions proportionate to the evidence.
Reproducibility and code quality	10	Fresh-environment execution, organised notebook, recorded dependencies and seeds, accessible assets, and regenerated report results.
Communication and professional practice	10	Coherent report and demonstration; page, time, and face requirements met; appropriate citations; complete AI and team disclosures.
Total	**100**	**Combined assessment of all three submission items.**

Submissions *more than five days late* or *incomplete submissions* receive **zero marks**.

Final submission check

Confirm that the notebook runs, the report is no longer than five pages, the video is no longer than three minutes and shows the presenter's face, all required figures regenerate, and AI and team statements are included. Check that uploaded files and checkpoint links are accessible to the marker.

Research-led learning and publication option

Contact the course instructor Dr Erick Purwanto for support in refinement and publication after the deadline for interested students or teams.

References

Alshehhi, A., Han, Z., & Anwer, R. M. (2026). *AgroGround: Multi-granularity grounded recognition in agriculture* [Preprint]. arXiv. https://arxiv.org/abs/2610.04425

Chilamkurthy, S. (2025, January 27). *Transfer learning for computer vision tutorial*. PyTorch. https://docs.pytorch.org/tutorials/beginner/transfer_learning_tutorial.html

Chai, A. Y. H., Lee, S. H., Tay, F. S., Bonnet, P., & Joly, A. (2024). Beyond supervision: Harnessing self-supervised learning in unseen plant disease recognition. *Neurocomputing, 610*, Article 128608. https://doi.org/10.1016/j.neucom.2024.128608

Dhamija, A. R., Günther, M., & Boult, T. E. (2018). Reducing network agnostophobia. *Advances in Neural Information Processing Systems, 31*. https://arxiv.org/abs/1811.04110

Dong, J., Fuentes, A., Zhou, H., Jeong, Y., Yoon, S., & Park, D. S. (2024). The impact of fine-tuning paradigms on unknown plant diseases recognition. *Scientific Reports, 14*, Article 17900. https://doi.org/10.1038/s41598-024-66958-2

Liu, W., Wang, X., Owens, J. D., & Li, Y. (2020). Energy-based out-of-distribution detection. *Advances in Neural Information Processing Systems, 33*, 21464–21475. https://arxiv.org/abs/2010.03759

Vaze, S., Han, K., Vedaldi, A., & Zisserman, A. (2022). *Open-set recognition: A good closed-set classifier is all you need?* International Conference on Learning Representations. https://www.robots.ox.ac.uk/~vgg/research/osr/

Wei, T., Chen, Z., Yu, X., Chapman, S., Melloy, P., & Huang, Z. (2025). *A large-scale in-the-wild dataset for plant disease segmentation* (Version v7) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.17719108

Wei, T., Chen, Z., Yu, X., Chapman, S., Melloy, P., & Huang, Z. (2026). A large-scale in-the-wild dataset for plant disease segmentation. *Scientific Data, 13*, Article 205. https://doi.org/10.1038/s41597-025-06513-4

Yan, S., Lou, L., Lin, J., Cernava, T., Migheli, Q., Oufensou, S., Chen, X., & Zhang, X. (2026). PAG-MSBNet: Prototype-aware gated multi-scale branch network for open-set crop disease recognition. *Ecological Informatics, 95*, Article 103703. https://doi.org/10.1016/j.ecoinf.2026.103703